Preparation Guide for Session 5: Deploying to Google Cloud

Deadline: Please complete this **before** our Session 5 class begins.

Hello everyone. This is a **MANDATORY** preparation guide you must complete before we start Session 5.

We received feedback from past students that the online lab can be fast-paced and hard to follow. This document is designed to provide you with everything you need *before* class, helping you understand our goals and be fully prepared for the hands-on lab. Our goal is to make sure no one gets lost or misses a step.

Part 1: What Will We Learn in Session 5?

In this session, we will take the most important step in MLOps: **deploying our API application to the cloud.**

We will transform the application that only runs on your local machine (localhost) into a public web service that anyone in the world can access.

Learning Objectives:

1. **Understand Cloud Architecture:** We will compare the two most common deployment methods: **Virtual Machine (VM)** (like renting an entire computer) and **Serverless** (like paying only for the code you run).
2. **Understand the "Build, Push, Deploy" Workflow:** This is the industry-standard process:
    - **Build:** Build a Docker image (we did this in Session 4 and will refine it).
    - **Push:** Push our image to a public "container registry" (specifically, **Docker Hub**).
    - **Deploy:** Tell Google Cloud to get the image from Docker Hub and run it as a public service using **Google Cloud Run**.
3. **Solve the MLflow Problem:** We will discuss how to handle our model (model.joblib) now that the MLflow server is no longer running alongside our API.

Hands-On Lab Goals:

You will personally type the commands to:

1. Update the Dockerfile to "bake-in" our model.
2. docker build and docker push your image to Docker Hub.
3. Configure and deploy that image to Google Cloud Run.
4. Receive a **public** URL and use it to test your live API.

Part 2: (MANDATORY) Preparation Checklist

You must complete all of these items **before our class begins**.

1. **Google Cloud Account (Most Important):**
    - Complete **Part 3** of this guide to sign up for a Google Cloud account and activate your **$300 in free credits**.
    - This is mandatory. You will not be able to do the lab without this.
2. **Docker Hub Account:**
    - Create a docker hub account.
    - Please make sure you **remember your username and password**. (e.g., your-dockerhub-username). We will use the docker login command in class.
3. **Your Session 4 Project:**
    - Make sure you have all your source code from Session 4 ready on your computer.
    - This includes: api.py, Dockerfile, requirements.txt.
    - We will continue working directly in this project folder.

Part 3: Google Cloud Setup (Homework)

This section guides you through signing up for Google Cloud. Please follow it carefully.

What You Will Need:

- **A Google Account:** Any Gmail or Google Workspace account.
- **A Credit/Debit Card:**
    - This is required by Google for identity verification (to prevent bot abuse).
    - **You will NOT be charged.** You will be using the Free Trial ($300).
    - We will also set a $1 budget alert so you can have total peace of mind.

Step 1: Sign Up for the Google Cloud Free Trial

You will receive **$300 in free credits** to use over 90 days. This is more than enough for our entire course.

1. Go to the Google Cloud Free Trial page: https://cloud.google.com/free
2. Click the "**Get started for free**" button.
3. Sign in with your Google account.

Step 2: Activate Your Account & Billing

Follow the on-screen prompts. This is a two-step process.

1. **Account Information:**
    - Select your country.
    - Agree to the Terms of Service.
2. **Payment Information:**
    - **Account Type:** Select **"Individual"**.

- **Payment Method:** Enter your credit/debit card information.
- **Why the card?** As mentioned, Google uses this to verify you are a real person. They will *not* charge your card for usage within the free tier.
- Click the "**Start my free trial**" button. You should see a welcome message and be redirected to the Google Cloud Console.

Step 3: Create Your First Project

All work in GCP is organized into "Projects." GCP may have created "My First Project" for you, but let's create a specific one for this course.

1. In the top-left corner of the console, click the project selection dropdown (it might say "My First Project").
2. In the pop-up window, click "**New Project**".
3. **Project Name:** Give it a clear name, like MLOps Course.
4. **Location/Organization:** You can leave this as "No organization."
5. Click "**Create**".
6. It may take a moment. Once created, make sure you **select your new MLOps Course project** from the dropdown menu at the top of the screen.

Step 4: (CRITICAL) Set a Budget Alert

This is your safety net. We will create a "budget" that *alerts* you if you ever start spending money (even the free credits). This is the single best way to ensure you never have a surprise bill.

1. In the top-left navigation menu (the "hamburger" icon ☰), go to **Billing**.
2. In the left-hand menu, select **Budgets & alerts**.
3. Click "**Create budget**".
4. **Name:** Give it a name like My $1 Alert.
5. **Projects:** Select your MLOps Course project.
6. **Amount:**
   - **Budget Type:** Select "**Specified amount**".
   - **Target Amount:** Enter 1 (for $1.00).
7. **Actions:**
   - Under **Email alerts**, check the box for "**Email alerts to billing admins and users**".
   - You can leave the other settings as default.
8. Click "**Finish**".

You're done! This budget **does not** turn off your services, but it will send you an email if your spending *ever* goes over $1. This gives you total peace of mind.

Part 4: Conclusion

You are now 100% ready for Session 5. You have your GCP account, free credits, and a safety alert in place. We'll see you in class!